# Colour-blind Series: Tritanopia

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## What is it for?

Tritanopia is a form of colour blindness that affects the ability to see blue. Protanopia and deuteranopia are both types of "red-green colour blindness". Tritanopia is different: it affects blue and yellow. People with tritanopia struggle to distinguish blue from green, and yellow from pink or red. It is significantly rarer than the red-green types, affecting less than 0.01% of the population.

In practice, this means colour choices that feel safe for red-green colour blindness can still cause problems. A design that uses blue and green to tell groups apart can be very hard to understand. So can yellow text on white, or white text on yellow.

## How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

## Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)